

# DENVar: A Global Dynamic Estimate of NDVI Variance

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## Abstract

Remote-sensing proxies of habitat productivity such as the Normalized Difference Vegetation Index (NDVI) have helped develop and support many hypotheses and projects. However, historically, users tended to focus solely on trends in mean productivity. In recent years, the variance around the mean (i.e., stochasticity) has gained more attention, particularly as climate change and human-induced rapid environmental change transform the ecosystems species have evolved in and adapted to. Still, calculating trends in NDVI stochasticity requires careful modeling, since stochasticity estimates are conditional on the estimated mean and the scale at which the data are collected and modeled. We thus present a new global Dynamic Estimate of NDVI Variance, *DENVar*. We estimate spatiotemporal trends in NDVI mean and variance using Generalized Additive Models, which provide a flexible yet transparent and interpretable model structure. We show that *DENVar* varies nonlinearly with **XXX** and can be used to test many hypotheses related to environmental stochasticity, such as optimal foraging theory and regime shifts as well as for conservation. We provide a publicly and freely available repository for downloading rasters of daily and long-term averages of *DENVar* and the related estimates of mean NDVI.

# 1 Introduction

The Normalized Difference Vegetation Index (NDVI) is a proxy of primary productivity used in a wide range of disciplines, including ecology (Pettorelli et al., b,a), landscape phenology (Merkle et al.; Hmimina et al.), agriculture (Franch et al.), forestry (João et al.), and conservation (Mallegowda et al.; Marcus et al.). High-resolution, remotely-sensed NDVI data have allowed analysts to estimate plant biomass without fieldwork (Bhatti et al.; Borowik et al.) and leverage such estimates for studying large-scale plant dynamics (Pettorelli et al., b) and their effects on animal movement behavior (Merkle et al.; Aikens et al., a). Still, most NDVI work is limited to quantifying changes in and responses to mean NDVI while ignoring changes in the variance around the mean.

Environmental stochasticity has been studied for decades (Levins; Stephens and Charnov; Lande; Sæther), but recent work has drawn more attention to the effects environmental stochasticity has at a wide range of scales, from individual organisms (Mezzini et al.; Rizzuto et al.) to landscapes (Zhang et al.; Mutti et al.; Marcus et al.). However, estimating trends in NDVI variance requires more nuanced modeling than estimating changes in mean NDVI, particularly when environments are temporally dynamic and spatially heterogeneous. The scale at which processes are observed determine how individuals' perception of and ability to respond to change (Levin). A forest fire is severely disruptive and deadly for many insects (Fettig et al.), while birds and large mammal may not struggle to escape the fire and or establish a new range (Franklin et al.; Calhoun et al.). Similarly, the effects of the fire recovery will depend on its duration relative to other organisms' phenology and lifespan (Frankham and Brook). Therefore, distinguishing between a change in mean NDVI and a stochastic fluctuation requires a conscious decision about the scale at which to consider the data.

Previous work suggests that environmental stochasticity can reduce limit species' ability to adapt or specialize (Chevin et al.), alter species' movement behavior (Mezzini et al.), and even affect extinction risk at both the population and species levels (Lande; Bull et al.; Fung et al.). Still, the effects of environmental stochasticity on species dynamics and ecosystem stability remain understudied (Marcus and Noonan). Global, temporally dynamic estimates of NDVI variance would facilitate research by building upon the widespread use and interpretability of NDVI while also removing the barriers of computational demand and statistical knowledge that are required for quantifying environmental stochasticity. However, since NDVI is not a direct measurement primary productivity but a proxy, it should be interpreted carefully (Pettorelli et al., a). NDVI is prone to sensitivity loss in very green environments (Matsushita et al.; Son et al.), values vary across sensors (Beck et al.; Fan and Liu; Huang et al.), and the coarse spatiotemporal scale of satellite-derived NDVI can be a substantial barrier when assessing fine-scale trends (Goodchild and Quattrochi; Reid et al.). However, careful statistical modeling can help overcome some of these limitations. In particular, Generalized Additive Models (GAMs) are particularly convenient due to their ability to learn flexibly from data (Wood; Simpson) and handle very large spatiotemporal datasets (Wood et al., a,b).

We present a new measure of environmental stochasticity, which we name the Dynamic Estimate of NDVI Variance (DENVar). We calculate DENVar using 44 years of NDVI data from the AVHRR and VIIRS collection of sensors and GAMs. Although satellites with AVHRR and VIIRS sensors

provide data at a coarser spatial resolution ( $0.05^\circ$ ) than other satellites (e.g., 250m for MODIS), they are the only sensors to provide data at a daily frequency and for more than 40 years. Our use of GAMs provides a flexible yet interpretable approach for estimating trends in mean and variance in NDVI without the rigidity of other more traditional methods (e.g., ARIMA) or the opaqueness of machine learning models such as neural networks (Simpson; Wood). We illustrate the relationships between DENVar and multiple, relevant ecological metrics, and we provide access to a free repository of daily, global estimates of DENVar and the corresponding estimates of mean NDVI.

## 2 Methods

### 2.1 Choice of color palettes

We represent NDVI using a modified version of Fabio Crameri’s divergent *bukavu* palette (Crameri, a; Crameri et al.), which has high-contrast for deuteranope and protanope vision. For NDVI values between -0.1 and 1, the colors also have sufficient contrast for the colors to also be distinguishable by strong tritanope and achromatic vision. Fig. A.1 contains an approximate representation of how each vision type would perceive each color palettes used. We obtained all color palettes with three or more colors from the *khroma* package (v. 1.14.0 Frerebeau) for R (v. 4.4.0; R Core Team).

### 2.2 Datasets used

Fig. 1 illustrates the processes of data collection, cleaning, and modeling used in this project. We downloaded NDVI image composites directly from the servers for data from AVHRR sensors (1981-06-24 to 2013-12-31; United States National Centers for Environmental Information (NCEI) and United States National Aeronautics and Space Administration; Vermote and NOAA CDR Program, a) and VIIRS sensors (2014-01-01 to 2025-06-29; United States National Aeronautics and Space Administration; Vermote and NOAA CDR Program, b).

We removed all pixels labeled as "cloudy" for AVHRR rasters and "probably cloudy" or "confidently cloudy" for VIIRS rasters. To remove pixels that included large portions of water, we used a shapefile of major water bodies (Garmin International, Inc. and Esri) to produce a static raster of the percentage of water within each pixel, at the same spatial resolution as the NDVI rasters ( $0.05^\circ$ ; Fig. A.2). We removed all pixels with  $> 40\%$  water cover.

Many rasters had horizontal bands of  $\text{NDVI} > 0.9$  at unrealistically northern latitudes between September and March, likely due to limited visibility during fall and winter in the Northern hemisphere. We identified and removed problematic pixels using the monthly empirical cumulative distribution functions of NDVI across latitude (Fig. A.3). We did not filter problematic values at latitudes  $< 60^\circ\text{S}$  since we fully excluded Antarctica from the analyses (Fig. A.4).

To ensure the number of rows for each dataset was below the maximum data frame size in R ( $3.17 \times 10^{10} > 2^{31} - 1 \approx 0.21 \times 10^{10}$ ), we reduced the dataset size by splitting it across four spatial groups (Figs. A.4 and A.5) and spatially and temporally aggregated the rasters by a factor of four. The final dataset thus had one raster every four days with a spatial resolution of  $0.05^\circ \times 4 = 0.2^\circ$  (Fig. A.6). Finally, we annotated the dataset with columns of day of year (1-366), integer year

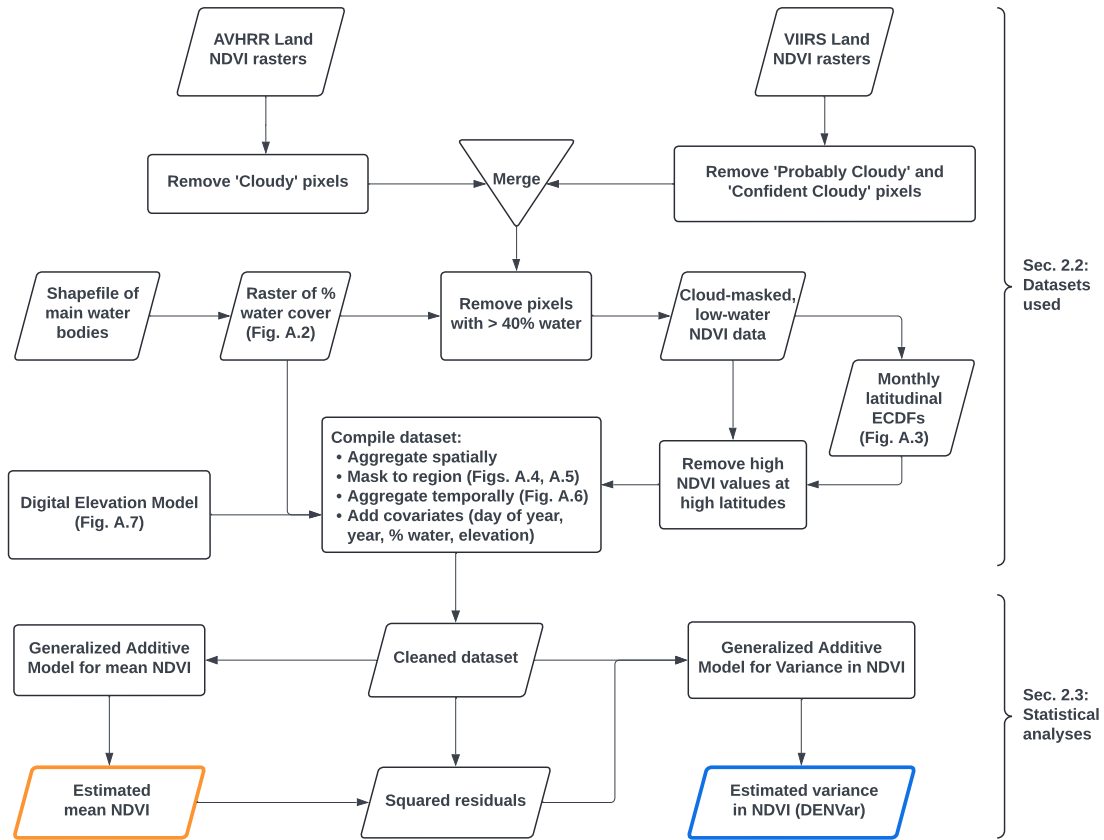


Figure 1: **Flowchart for the analyses used in the manuscript.** The curly braces on the right indicate the subsections in the Methods section that detail the dataset collation and cleaning (2.2), and statistical analyses (2.3).

(1981-2025), percent water cover, and elevation above sea level (m). The digital elevation model was obtained using the `get_elev_raster()` function from the `elevatr` package for R (v. 0.99.0; Hollister et al.) with an original resolution of  $0.076^\circ$  (Fig. A.6). Despite the substantial reduction in dataset size, the dataset creation was still challenging for the machine used, which had 2.2 TB of RAM and two Intel Xeon Platinum 8462Y+ processors (64 cores total).

## 2.3 Statistical Analyses

We estimated spatiotemporal trends in mean and variance in NDVI using Generalized Additive Models (GAMs) via the `mgcv` package for R (v. 1.9-3 Wood). To reduce modeling fitting times with negligible losses to model accuracy, we used the `bam()` function with fast REstricted Marginal Likelihood (`method = fREML`) and covariate discretization (`discrete = TRUE` Wood et al., a,b). Ideally, NDVI should be modeled using beta location-scale models (after the linear transformation  $Y^* = (Y + 1)/2$ ) to account for the fact that: (1) NDVI is bounded between -1 and 1, and (2) the variance in NDVI is dependent on the mean (and vice-versa), since ecosystems with very low NDVI

(e.g., rock, ice, or concrete) or very high NDVI (e.g., dense forest) tend to have lower variance in NDVI. However, preliminary tests and work by (Marcus et al.) showed that beta GAMs and particularly beta location-scale GAMs had prohibitive requirements for both fitting times and memory requirements, since both requirements scale  $O(Mnp^2)$ , for  $M$  smoothing parameters and a model matrix with  $n$  rows and  $p$  columns (Wood et al., a). In contrast, Gaussian models with fixed scaled parameters (i.e., constant variance) fit substantially faster and provided very similar spatiotemporal estimates of mean and variance in NDVI (Appendix B). Thus, we estimated trends in mean and variance in NDVI using a pair of GAMs for each region shown in Fig. A.4; one for estimating mean NDVI over space and time (assuming a constant variance), and one for estimating the mean squared residual (i.e., variance) in NDVI (again, assuming a constant scale parameter).

Together, the sets of models for mean NDVI ( $\mu$ ) and variance in NDVI ( $\sigma^2$ ) had the following structure:

$$\left\{ \begin{array}{l} \text{NDVI} \sim \text{Normal}(\mu, \sigma^2) \\ \mu = \beta_{\mu} + s(\text{prop\_water}) + s(\text{elev\_m}) + s(\text{doy}) + s(\text{year}) + s(\text{lat, long}) + \\ \quad t(\text{doy, year}) + t(\text{doy, lat, long}) + t(\text{year, lat, long}) \\ \sigma^2 = \beta_{\sigma^2} + s(\text{prop\_water}) + s(\text{elev\_m}) + s(\text{doy}) + s(\text{year}) + s(\text{lat, long}) + \\ \quad t(\text{doy, year}) + t(\text{doy, lat, long}) + t(\text{year, lat, long}) + s(\mu) \end{array} \right. , \quad (1)$$

where, for a given (spatiotemporally aggregated) pixel, **prop\_water** is the proportion of water within the pixel, **elev\_m** is elevation above sea level in meters, **doy** is integer day of year (1-366), **year** is integer year (1981-2025), **lat** and **long** are latitude and longitude, respectively. For ease of readability, we omit the pixel-level indices.  $s(\ )$  indicates marginal smooth functions, while  $t(\ )$  indicates tensor product interaction terms. Seasonal terms used cyclical cubic splines (**bs** = 'cc') with edge knots 0.5 and 366.5, while spatial smooths used a basis with second-order spline on the sphere, computed by Wendelberger's (wen) method (**bs** = 'sos'; Wood). The linear predictors for  $\mu$  and  $\sigma^2$  have the same structure, other than the  $f_9(\mu)$  smooth term for  $\sigma^2$ , but the functions for  $\sigma^2$  are independent of those for  $\mu$ ; we use the same in. Despite the data aggregation procedures and faster fitting methods, including parallelizing computation across 60 cores, each of the four models required multiple days to fit.

The residuals from each of the four models served to estimate the spatiotemporal trends in the variance. However, the discretized predictions from models 1 were excessively coarse over space, so predictions were calculated without discrete prediction methods (**discrete** = **FALSE**), at the cost of substantially longer computational times. A second set of GAMs estimated the variance as the mean squared residual across time and space using the model structure as above and an additional smooth term of the estimated mean NDVI:

where smooth terms are defined as with model 1 but are independent from it, and  $f_9(\mu)$  indicates a smooth term of mean NDVI for a given location in space and date in time. It is worth noting

that, since there are no corrections to any bias in residuals from models 1, the estimates of variance from models 1 may be inflated near areas or periods of rapid and substantial change. Still, the appreciability of this inflation is dependent on the spatiotemporal scale over which changes are perceived (Levin; Frankham and Brook; Goodchild and Quattrochi; Mezzini et al.).

From the two sets of models, we produced daily estimates of (1) the Dynamic Estimate of NDVI Variance (DENVar) and (2) the estimated mean NDVI they are conditional on, at a **XXX**°spatial scale. The rasters are freely available on GitHub (see code availability statement), along with long-term (1981-2025) averages of both DENVar and the estimated mean NDVI. Since estimates of variance are conditional on the corresponding estimates of the mean, we suggest users always use the corresponding rasters of mean NDVI in their analyses rather than using independent estimates of mean NDVI.

### 3 Results

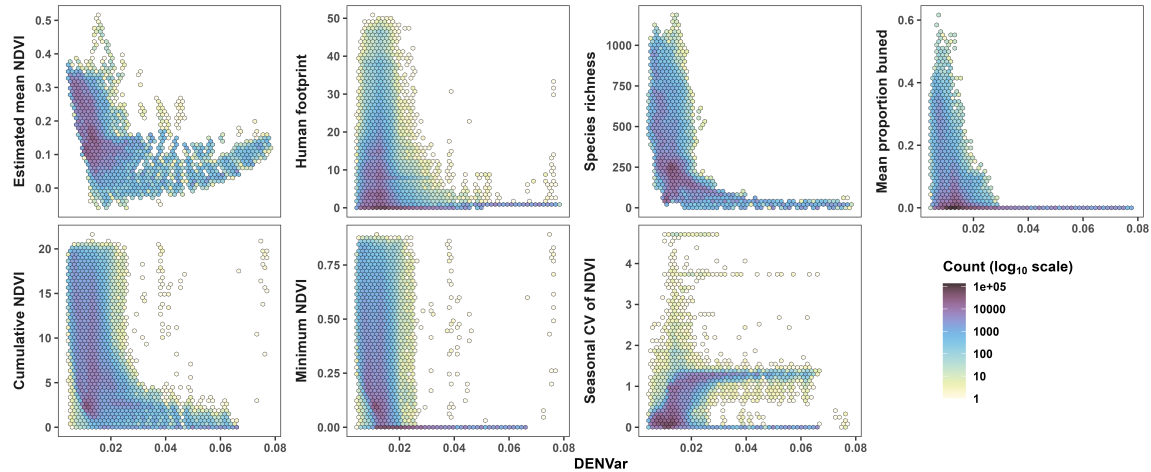
Figures:

- two Robinson projections of the world: long-term mean NDVI and DENVar on spring solstice OR the world with ecozone shapefiles to see if fill matches ecozone boundaries (two columns)
- hex plots of pred vs obs for mean and var (one column)
- surface plots of year (x) and latitude (y) for mean NDVI and DENVar (to detect trends over space and time; two columns)
- hex plots of DENVar with other variables (add temperature, annual precip, extreme weather event, fires, invasive species, max body size, disease occurrence, ...; two columns)
- supplementary material – appendix C (outputs): histograms of mean around flat map of biomes (already created, but change colors?)

#### 3.1 Correlations with other metrics

- temperature
- seasonal temperature range
- precip
- seasonal temperature range
- spp diversity
- gross primary productivity
- max animal weight
- <https://silvis.forest.wisc.edu/globalwui/>

- <https://silvis.forest.wisc.edu/data/dhis/>: NDVI16 (Normalized Difference Vegetation Index), EVI16 (Enhanced Vegetation Index), FPAR8 (Fraction absorbed Photosynthetically Active Radiation), LAI8 (Leaf Area Index), GPP8 (Gross Primary Productivity)



## 3.2 Obtaining the data

- Data should be available as daily rasters on a github repo?
- make an app for data annotation?

## 4 Discussion

We have presented a new spatiotemporally dynamic metric of environmental stochasticity, which we have called the Dynamic Estimate of NDVI Variance, DENVar. In this section, we discuss the interpretation of DENVar in greater depth, and we argue the advantages and limitations of the statistical methods used. We conclude by suggesting some use cases of DENVar within animal movement ecology, landscape ecology, and phenology, but we invite readers to consider how DENVar may be of use in their field and adjacent disciplines.

### 4.1 Interpreting DENVar: the importance of scale

difficulties in estimating variance

what counts as a change? predictable vs stochastic changes

smoothness of mean impacts smoothness and size of variance: depends on scale of interest (e.g., animals' ability to respond to, learn, and predict changes in the mean). Changes not attributed to trends in the mean are stochastic

DENVar is a good metric as long as the spatiotemporal scale of interest is similar or larger to that of DENVar. May not work well for small-scale processes such as the movement dynamics of a small rodent, the phenology of a small orchard when spatiotemporal heterogeneity is finer than DENVar’s resolution

## 4.2 Environmental stochasticity and animal movement ecology

The field of animal movement ecology has long shown that animals alter their movement in response to resource abundance or availability (Burt; Southwood; Charnov; Broekman et al.). More recently, movement ecologists have started focusing more on resource stochasticity (Stephens and Charnov; Rizzuto et al.; Marcus and Noonan), but it is worth distinguishing between resource stochasticity and seasonality (Nilsen et al.). Seasonality is generally best interpreted as a cyclical change in the mean that many animals can learn to predict, as in the case of green-wave surfing (Middleton et al.; Geremia et al.). In contrast, environmental stochasticity refers to unpredictable changes, such as forest fires [ref](#), sudden snow storms (Aikens et al., b), and unusual phenology [ref](#). Animals also respond to stochastic changes (Mezzini et al.), but their ability to adapt depends on behavioral plasticity (Steixner-Kumar and Gläser; Rickbeil et al.), and memory (Polansky et al.; Abrahms et al.; Riotte-Lambert and Matthiopoulos; Falcón-Cortés et al.; Ranc et al.).

evolutionary tipping points (Botero et al.)

multifarious change (Polazzo et al.)

extreme events, black swans (Anderson et al.; Logares and Nuñez)

- migration routes are key habitat (Ortega et al., in prep), especially in stochastic environments

- can fail to adapt to changes in conditions (Sawyer et al.)

- <https://www.sciencedirect.com/science/article/pii/S0960982220308484>

- <https://onlinelibrary.wiley.com/doi/full/10.1111/gcb.15169>

- industrial development affects animals’ ability to respond to both predictable and unpredictable change (<https://www.nature.com/articles/s41559-022-01887-9>)

- effects of sudden deep snow in red desert on proghorn (<https://www.sciencedirect.com/science/article/abs/pii/S0960982220308484>)

- <https://www.nature.com/articles/s41467-023-37750-z>

- <https://esajournals.onlinelibrary.wiley.com/doi/full/10.1002/ecy.4238>

- detecting regime shifts (in lakes: (Bjorndahl et al.))

## 4.3 Environmental stochasticity and population dynamics

read lande\_risks\_1993

Environmental stochasticity reduces habitats’ energetic balance and carrying capacity (Chevin et al.) as individuals struggle to rely on unpredictable resources and increase the instability of populations

231 and communities (Lande). Additionally, stochastic environments reduce the ability of specialist  
232 species to adapt and evolve while selecting for more generalist species (Levins). Consequently,  
233 stochastic environments tend to have lower species diversity (Marcus et al.).  
234 Recent work by Devries has suggested that NDVI stochasticity is an important metric for evaluating  
235 forest recovery after large fires, while Collinson has suggested that variance in NDVI is an important  
236 predictor of the occurrence of lightning-induced forest fires.

- 237 • jonzen\_rapid\_2006
- 238 • effects can be compounded by cumulative effects (e.g., loss of species diversity; (Wolf et al.))
- 239 • climate change affects plant phenology, but changes in situ may not match experimental esti-  
240 mates (Wolkovich et al.)
- 241 • reproduction of painted lady butterfly (Stefanescu et al.)
- 242 • wessling\_seasonal\_2018

#### 243 4.4 Environmental stochasticity and fire ecology

#### 244 4.5 Environmental stochasticity and phenology

### 245 5 Other useful references

- 246 • **READ**: Methods for Detecting Early Warnings of Critical Transitions in Time Series Illustrated  
247 Using Simulated Ecological Data (<https://doi.org/10.1371/journal.pone.0041010>)
- 248 • **READ**: <https://doi.org/10.1029/2024GL109298>
- 249 • **READ**: Site fidelity as a maladaptive behavior in the Anthropocene: <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecs.25888>
- 250 • **READ**: Pettorelli N., Vik J.O., Mysterud A., Gaillard J.-M., Tucker C.J. & Stenseth N.Chr.  
251 (2005). Using the satellite-derived NDVI to assess ecological responses to environmental  
252 change. Trends in Ecology & Evolution 20, 503–510. <https://doi.org/10.1016/j.tree.2005.05.011>
- 253 • Ecological and evolutionary consequences of changing seasonality: <https://www.science.org/doi/10.1126/science.1274441>
- 254 • increase in variance before eutrophication in lakes: <https://doi.org/10.1002/lno.10355>
- 255 • uses ndvi variance: <https://esajournals.onlinelibrary.wiley.com/doi/full/10.1002/ecm.1348>
- 256 • bro-jorgensen\_using\_2008: in the dry season preceding the rut topi antelope (*Damaliscus*  
257 *lunatus*) respond to forage availability at the large scale but not the fine scale
- 258 • keith\_predicting\_2008
- 259 • chevin\_adaptation\_2010
- 260 • rickbeil\_plasticity\_2019
- 261 • mueller\_search\_2008
- 262 • pettorelli\_using\_2005
- 263 • nilsen\_can\_2005
- 264 • merkle\_large\_2016
- 265 • tian\_evaluating\_2015
- 266 • huang\_commentary\_2021
- 267 • fan\_global\_2016
- 268 • wang\_stochastic\_2019
- 269 • pease\_ecological\_2024
- 270 • xu\_plasticity\_2021
- 271 • calibrating across sensors: <https://ieeexplore.ieee.org/abstract/document/6809983>

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### 275 General

276 Thank others for any contributions, whether it be direct technical help or indirect assistance

### 277 Author Contributions

278 Describe contributions of each author to the paper, using the first initial and full last name.

279 Examples:

280 “S. Zhang conceived the idea and designed the experiments.”

281 “E. F. Mustermann and J. F. Smith conducted the experiments.”

282 “All authors contributed equally to the writing of the manuscript.”

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284 Name financially supporting bodies (written out in full), followed by the funding awardee and asso-  
285 ciated grant numbers (if applicable) in square brackets.

286 Example:

287 “This work was supported by the Engineering and Physical Sciences Research Council grant  
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289 Research Project Grant.”

290 If the research did not receive specific funding, but was performed as part of the employment  
291 of the authors, please name this employer. If the funder was involved in the manuscript writing,  
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302 was employed or paid by a funder, in the commissioning, conception, planning, design, conduct, or  
303 analysis of the work, the preparation or editing of the manuscript, or the decision to publish must  
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307 declaring their interests. Declared conflicts of interest will be considered by the editor and reviewers  
308 and included in the published article.

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310 A data availability statement is compulsory for all research articles. This statement describes  
311 whether and how others can access the data supporting the findings of the paper, including 1)  
312 what the nature of the data is, 2) where the data can be accessed, and 3) any restrictions on data  
313 access and why.

314 If data are in an archive, include the accession number or a placeholder for it. Also include any  
315 materials that must be obtained through a Material Transfer Agreements (MTA).

## 316 **Supplementary Materials**

317 Describe any supplementary materials submitted with the manuscript (e.g., audio files, video clips  
318 or datasets).

319 Please group supplementary materials in the following order: materials and methods, figures,  
320 tables, and other files (such as movies, data, interactive images, or database files).

321 Example: Fig. S1. Title of the first supplementary figure.

322 Fig. S2. Title of the second supplementary figure.

323 Table S1. Title of the first supplementary table.

324 Data file S1. Title of the first supplementary data file.

325 Movie S1. Title of the first supplementary movie.

326 Be sure to submit all supplementary materials with the manuscript and remember to reference  
327 the supplementary materials at appropriate points within the manuscript. We recommend citing  
328 specific items, rather than referring to the supplementary materials in general, for example: “See  
329 Figures S1-S10 in the Supplementary Material for comprehensive image analysis.”

330 A link to access the supplementary materials will be provided in the published article.

331 Supplementary Materials may include additional author notes—for example, a list of group  
332 authors.

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